# DELIGNE-MUMFORD STACKS WITHOUT AUTOMORPHISMS ARE EXACTLY THOSE REPRESENTABLE BY AN ALGEBRAIC SPACE

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*Editorial note.* No omission markers in-span. Same-chapter inputs not carded are declared external.

*Allowed standard prerequisites (editorial).*

- Setoid characterizations. Stacks, lemmas characterize-stack-in-setoids and stack-in-setoids-characterize (cross-chapter, named). Give (1) iff (2) and the equivalence j: X -> S_F.
- Representable implies DM. lemma-representable-algebraic (carded L26682). Gives (3) implies (1).
- Diagonal transfer to presheaves. lemma-map-presheaves-representable-by-algebraic-spaces (same chapter, NOT carded, declared external). Transfers representability of Delta_X to Delta_F.
- Setoid property transfer. lemma-map-fibred-setoids-property (same chapter, NOT carded, declared external). Concludes h_U -> F is surjective etale.
- Bootstrap theorem. Bootstrap, theorem bootstrap (cross-chapter, named). Final step: F is an algebraic space.

*Source reference.* Original reference 03Y2: focused-seeds/geometry/S9-alg-algebraic-stack-no-automorphisms/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03Y3: focused-seeds/geometry/S9-alg-algebraic-stack-no-automorphisms/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04TC: sources/stacks-algebraic/pinned-original-source.tex, source lines 1138–1138. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0458: sources/stacks-algebraic/pinned-original-source.tex, source lines 839–839. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 03YS: sources/stacks-algebraic/pinned-original-source.tex, source lines 1682–1682. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04ZM: focused-seeds/geometry/S9-alg-algebraic-stack-no-automorphisms/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0432: focused-seeds/geometry/S9-alg-algebraic-stack-no-automorphisms/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme contained in $Sch_{fppf}$. Let $\mathcal{X}$ be an algebraic stack over $S$. The following are equivalent*

(1) *$\mathcal{X}$ is a Deligne-Mumford stack and is a stack in setoids,*
(2) *$\mathcal{X}$ is a Deligne-Mumford stack such that the canonical 1-morphism $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is an equivalence, and*
(3) *$\mathcal{X}$ is representable by an algebraic space.*

*Proof.* The equivalence of (1) and (2) follows from Stacks, Lemma 04ZM. The implication (3) $\Rightarrow$ (1) follows from Lemma 03YS. Finally, assume (1). By Stacks, Lemma 0432 there exists a sheaf $F$ on $(Sch/S)_{fppf}$ and an equivalence $j : \mathcal{X} \to \mathcal{S}_F$. By Lemma 0458 the fact that $\Delta_\mathcal{X}$ is representable by algebraic spaces, means that $\Delta_F : F \to F \times F$ is representable by algebraic spaces. Let $U$ be a scheme, and let $x : (Sch/U)_{fppf} \to \mathcal{X}$ be a surjective étale morphism. The composition $j \circ x : (Sch/U)_{fppf} \to \mathcal{S}_F$ corresponds to a morphism $h_U \to F$ of sheaves. By Bootstrap, Lemma 03Y2 this morphism is representable by algebraic spaces. Hence by Lemma 04TC we conclude that $h_U \to F$ is surjective and étale. Finally, we apply Bootstrap, Theorem 03Y3 to see that $F$ is an algebraic space. □